

Cheng Dai

[[Email](#)] | [[Github](#)] | [[Homepage](#)] | [[Google Scholar](#)] | [Beijing, China]

OVERVIEW

I'm a researcher specializing in thermal sensing, video generation, and physical AI.

My research investigates how generative models can learn the dynamics, geometry, and physical structure of real-world environments, and how these models can support perception, planning, and interaction.

POSITIONS

Tsinghua University, Institute for AI Industry Research

Jul. 2026 - ?

Research Assistant

Supervisors: [Chaojian Li](#) and [Hao Zhao](#)

- Dissertation Fields: Quantum Machine Learning, and World Model.
- Collaborating with technical teams from BAAI, HKUST and QBoson Inc.

Westlake University, School of Science

Sep. 2025 - Jul. 2026

Research Assistant

Supervisors: [Xin Yuan](#) and [Fanglin Bao](#)

- Dissertation Fields: HADAR (nature2023, Temperature-emissivity-texture-distance decomposition)
- AI department manager in the flab. Conduct Research and guide students.
- Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, **HAIR**.
- Release the first temperature-emissivity-texture decomposition dataset and benchmark, **TeX-1500**.
- Release the first sparse-active thermal texture imaging framework that isolates reflection-dominant thermal texture through controlled LWIR illumination and leverages RGB-domain pretrained representations for reconstruction, **T²exture**.

EDUCATION

Wuhan University

Sep. 2025 - Mar. 2026

Graduate Study in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering (LIESMARS), Major in Communication and Information Systems

Supervisors: [Wei He](#)

- GPA: 90.79% (among 6 postgraduate courses). recipient of the First-Class Graduate Scholarship.
- Voluntarily withdrew after consultation with my supervisor to pursue a more suitable research direction in physical AI.

Jilin University

Sep. 2021 - Jun. 2025

BSc. Major in Communication Engineering

- GPA: 90.58%. Rank: 12/283 (Outstanding graduate, top 5%), 508/710 in CET6. (German and Russian study for 2 years)
- Outstanding Undergraduate Thesis (top 3%).
- Outstanding Student of the School (*4). First-Class Scholarship, Second-Class Scholarship (*3)

PUBLICATIONS

[1] **Cheng Dai**, Jiale Lin, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [[Paper](#)], [[Homepage](#)], [[huggingface](#)] (arxiv version, published in the future)

[2] **Cheng Dai**, Jiale Lin, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [[Paper](#)], [[Homepage](#)] (IEEE **TIP** under review)

[3] Jiashuo Chen, **Cheng Dai (co-first author and listed alphabetically)**, and Fanglin Bao

T2exture: Sparse Active Thermal Texture Imaging. [[Paper](#)], [[Homepage](#)], [[huggingface](#)] (**AAAI** under review)

Note: Independently propose the method, finish main experiments, draw main figures, and write the top-4 sections.

[4] Hongyi Xu, Jiale Lin, **Cheng Dai**, Du Wang, Chenjun Zhao, Jiashuo Chen, Liqin Cao, Yanfei Zhong, Yiyuan She, Fanglin Bao.

Universal computational thermal imaging overcoming the ghosting effect. [[Paper](#)] (**Optica** Under review)

INTERNSHIP

Lightwheel, Embodied Assets Group

Dec. 2024 - Feb. 2025

Group Leader: [Jian Yang](#), and [Soroush Nasiriany](#)

- Built tools for embodied simulation, asset construction, data collection, and teleoperation with Blender, MuJoCo, Robosuite, Issacsim and RoboCasa. [[Few Contribute Codes](#), Merged in Final [Robocasa](#)]
- Achievement: develop spacemouse, blender2yaml, yaml2blender plugins, reducing validations cycles 20× Faster.

Amoon AI, Large Audio Model Department

Apr. 2025 - Sep. 2025

- Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.
- Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

AWARDS IN COMPETITION

- First Prize in the Provincial Level of the National Undergraduate Mathematics Competition.
- Honorable Mention in the American Mathematical Competition.
- Student Leader in Science and Technology Innovation Plan (Changchun Institute of Optics, CAS), **excellent completion**.
- Student Leader in Provincial-level college student innovation and entrepreneurship competition, successful completion.

SKILLS

- Train and Finetune pretrained Generative Model (Language Model, Vision Language Model, Audio Model .etc.)
- Improvements to Generative Model (Multi-branch diffusion, Schrodinger Bridge or inference enhancement .etc.) and Interpretability Analysis (i.e. Fisher Information theory, Convex Theory, Derivation of differential equations).
- Quantum Calculation and Quantum Machine Learning.
- Model-based Physical Inversion, Numerical Computation, Hyperspectral Image Processing.
- Time Series Model (i.e. PatchTST, DLinear, TimesFM) and signal processing (i.e. EEG, ECG, Earth vein waves)

Software: [Cuda-Jax, Latex, Codex, Git, Ubuntu, Python, C++, FFmpeg, Html5, CSS3, Javascript, MATLAB].